

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

MODERN MANUFACTURING METHODS

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) What are the process parameters of ultrasonic machining process? | 2M |
| (b) What is Abrasive Water Jet Machining? | 2M |
| (c) What is Electrode? Give its application. | 2M |
| (d) Give the applications of Di-electric fluid. | 2M |
| (e) Applications of Electro Chemical Honing. | 2M |
| (f) What are various Techniques of Applying Maskants? | 2M |
| (g) What is Machining, give its applications? | 2M |
| (h) What are the sources of Thermal energy in manufacturing? | 2M |
| (i) Give applications of Sterolithography (SLA). | 2M |
| (j) Give any application of Laminated Object Manufacturing. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the need and characteristic features of non-traditional machining? What are the advantages of non-traditional machining process? 10M
- OR**
- 3 What is Jet machining process? Explain any two Mechanical energy-based Jet machining processes. 10M
- 4 What are the important parameters that control the material removal rate in electrical discharge machining? Briefly explain any two factors. 10M
- OR**
- 5 What is Wire cut EDM? Explain Working Principle and Applications of wire cut EDM. 10M
- 6 Difference between Chemical Machining and Electro Chemical Machining. Give their application individually. 10M
- OR**
- 7 Differentiate between Electro Chemical Grinding and Electro Chemical Honing. What are their advantages and limitations? 10M
- 8 How is the metal removal rate controlled in plasma arc machining process? Give a brief note on scope of the application and the process limitations of plasma arc machining process. 10M
- OR**
- 9 (a) What is the working principle of Electron Beam Machining? 5M
(b) Give applications, merits and demerits of electron beam machining of process. 5M
- 10 What is Additive manufacturing? Explain classifications of Additive Manufacturing Process? 10M
- OR**
- 11 Give brief explanation on Selective Laser Sintering and Fused Deposition modelling with applications? 10M
